A Credential Controlled Digital Library for Institutional Distribution of Electronic Reading Material

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System Design and Implementation Report

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Abstract. Institutional libraries serving students in bandwidth constrained environments face a distribution problem that neither physical collections nor commercial learning platforms resolve economically. A physical collection serves one reader per copy and imposes opening hours; commercial platforms impose per seat licensing that scales poorly for institutions with limited budgets. This report presents the design, implementation and evaluation of a digital library built entirely on managed services with no recurring platform cost at small scale. The system distributes electronic books to a closed population of enrolled students through an Android application, with cataloguing performed through a private administrative interface. Access is granted by credential issuance rather than self registration, and every catalogued title is available to every active student simultaneously. The implementation combines Firebase Authentication for identity, Cloud Firestore for the catalogue, Google Drive for file storage and a serverless application layer on Vercel. A direct browser to storage transfer path removes the request body limit that would otherwise restrict book size to 4.5 megabytes. Functional verification against the deployed system confirmed correct behaviour across fourteen test cases including authorisation boundaries, identity normalisation and delivery integrity. A capacity analysis projecting the collection from its current state to ten thousand titles identifies three constraints that bind before any storage limit is reached, of which unbounded catalogue reads is the most severe: at ten thousand titles, five client sessions exhaust the free daily read allowance. Remedies are quantified and sequenced.

Index Terms. digital libraries, controlled distribution, serverless architecture, offline first mobile applications, capacity planning, educational technology.

I. INTRODUCTION

The provision of reading material to enrolled students is a problem of distribution rather than of information retrieval. The titles are known, the readership is enumerable, and the requirement is simply that the right material reaches the right people at acceptable cost. Physical collections address this poorly at the margin: a single copy serves a single reader, demand concentrates around examination periods precisely when contention is highest, and duplicate acquisition is expensive and seasonal in its utility.

Commercial digital library platforms exist and are capable, but their commercial model is per seat licensing renewed annually. For an institution whose constraint is capital rather than technical capability, that model converts a one time cataloguing effort into a permanent operating liability.

This report documents a third approach: assembling a purpose built system from managed services whose free tiers accommodate the workload, accepting narrower functionality in exchange for the elimination of licensing cost. The contribution is not architectural novelty. Each component is used conventionally. The contribution is an account of which trades were made, why, and where the resulting design ceases to hold as the collection grows.

Section II establishes the operating context. Section III examines alternatives that were considered and rejected. Sections IV through VI describe the architecture, implementation and security model. Section VII reports functional verification against the deployed system. Section VIII presents a capacity analysis projecting the collection to ten thousand titles, which is the principal analytical contribution of this report. Sections IX and X discuss limitations and the work they imply.

II. PROBLEM CONTEXT

A. Operating conditions

Four conditions shaped the design and are stated explicitly because the design is not defensible without them.

Capital constraint. No recurring licence fee is affordable. This ruled out commercial platforms and directed the design toward services whose free allowances accommodate the expected workload.

Absence of technical staff. The institution employs no systems administrator. Any component requiring patching, backup scheduling or capacity provisioning would in practice go unmaintained. This ruled out self operated infrastructure irrespective of its cost advantages.

Intermittent and metered connectivity. Continuous streaming presumes reliable and affordable bandwidth. That presumption does not hold uniformly across the student body. The design therefore treats a network connection as required once per title and not thereafter.

Controlled readership. Material held by the institution must reach enrolled students and no one else, and access must terminate when enrolment does. This requirement is narrow but firm, and it excludes any design in which readers enrol themselves.

B. Derived requirements

From these conditions the following requirements follow directly. R1: no recurring platform cost at expected scale. R2: no institutionally operated server. R3: complete offline reading after a single download. R4: accounts created by an administrator, never by the reader. R5: immediate revocation of access on suspension. R6: cataloguing achievable by a person with no technical training.

III. ALTERNATIVES CONSIDERED

Four approaches were evaluated against the requirements of Section II. Table I summarises the comparison.

TABLE I. COMPARISON OF CANDIDATE APPROACHES AGAINST DERIVED REQUIREMENTS

Approach	R1 cost	R2 no server	R3 offline	R4 controlled	Determining factor
Expanded physical collection	Fails	Passes	Passes	Passes	Duplicate acquisition cost scales with contention, not with collection value
Commercial platform, per seat licence	Fails	Passes	Partial	Passes	Converts a one time effort into a permanent annual liability
Self operated server and object store	Partial	Fails	Passes	Passes	Requires maintenance capability the institution does not have
Managed services, purpose built client	**Passes**	**Passes**	**Passes**	**Passes**	**Selected. Cost transferred from licensing to a bounded cataloguing effort**

A. Storage service selection

Within the selected approach, the object store required a further decision. Firebase Storage integrates most naturally with the remainder of the platform but requires the metered Blaze plan, introducing a recurring charge from the first uploaded file and violating R1 [1]. Google Drive provides fifteen gigabytes at no cost on a consumer account and scales thereafter through consumer priced tiers [2]. The functional difference for file delivery is negligible. Drive was selected on cost.

This selection carries a consequence examined in Section VI: Drive has no native mechanism for short lived signed retrieval addresses, which constrains the delivery model available to the mobile client.

IV. SYSTEM ARCHITECTURE

The platform comprises five components across four managed services. No component is operated by the institution. Table II states the division of responsibility.

TABLE II. COMPONENT RESPONSIBILITIES

Component	Service	Responsibility
Identity	Firebase Authentication	Credential verification, session issuance, role assertion, immediate revocation
Catalogue	Cloud Firestore	Book records, student profiles, declarative access rules
File store	Google Drive	Book files and their retrieval addresses
Application	Vercel	Administrative interface and its serverless functions
Client	Android application	Catalogue browsing, download, offline reading

A. Transfer path

Serverless platforms bound the size of a request body. On the selected host that bound is 4.5 megabytes [3], which is smaller than the majority of the intended collection. Rather than accept a ceiling excluding most titles, the design routes file content past the application layer entirely.

The administrative client requests a transfer session. The application layer, holding the storage credential, opens a resumable session with the storage service [4] and returns the issued address. The client then transfers the file directly. Only on completion does the application record the catalogue entry. Figure 1 shows the sequence.

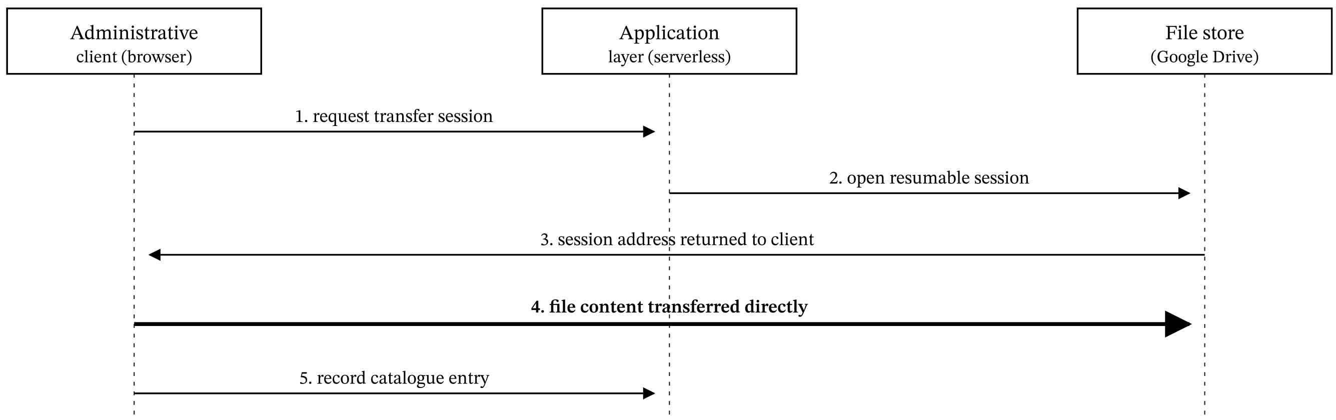

Fig. 1. Upload sequence. Step 4 carries the file content and does not traverse the application layer, which removes the request body limit as a constraint on book size. An advisory ceiling of 512 megabytes is applied at step 1 to reject accidental selection of an unintended file.

Failure handling is symmetric. If the transfer fails, no catalogue entry is created. If the catalogue write fails after a successful transfer, the stored file is deleted. Neither ordering can leave a stored file that no catalogue entry references.

B. Retrieval path

Stored files are marked readable by any bearer of their address. The mobile client therefore retrieves content without holding any storage credential, which permits the distributed application package to contain no secret. Discovering an address nonetheless requires an authenticated catalogue read. Section VI examines the limits of this arrangement.

V. IMPLEMENTATION

A. Identity normalisation

The identity service accepts electronic mail addresses as the sole identifier form. Requiring every student to hold a functioning mailbox would exclude some and delay others, violating R6 in spirit if not in letter.

The implementation therefore accepts an institutional roll number and maps it deterministically onto an address in a reserved domain that receives no mail. A student entering `2024-BS-014` authenticates as `2024-bs-014@ruthpuaf.local` and is never shown the translation. Normalisation folds case and collapses separator characters, so that `2024 / BS / 014` and `2024-BS-014` resolve identically and a duplicate is refused at creation. An administrator may alternatively issue a genuine address where one exists; the system accepts either form and records which was used.

B. Privilege separation

Account creation cannot occur in the administrative client. Invoking the client identity library to create a user would replace the caller's own session with that of the created account, signing the administrator out. Creation therefore occurs in the application layer using an administrative credential, which additionally permits the role assertion to be attached at creation. That assertion is carried inside the authentication token and cannot be altered by any client.

C. Offline reading and delivery integrity

A retrieved file is written to application private storage and opened from there subsequently, satisfying R3. The client verifies content before accepting it. This is necessary rather than defensive: the storage service returns a hypertext document in place of file content under several conditions, including its malware scanning interstitial for larger files and its access denial page, and both arrive with a success status. The client inspects leading bytes and rejects content beginning as markup, and rejects a document declared as PDF whose signature does not match. Without this check the client would store a web page and present it to the reader as a book.

> ***Implementation note***
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> The retrieval address form matters. The legacy address form serves the malware interstitial for files above approximately one hundred megabytes. The implementation uses the current form with explicit confirmation, which returns content directly.

VI. SECURITY MODEL

A. Enforcement

Authorisation is enforced at three independent layers. The interface layer signs out an account lacking the administrative role, which is a usability measure rather than a control. The application layer verifies the presented token and its role assertion on every mutating request, and this is the operative control. The database layer permits students to read the catalogue and their own profile only, and permits no client to write at all; every write occurs through the application layer using a credential that bypasses these rules by design.

B. Credential exposure

TABLE III. CREDENTIALS AND THEIR EXPOSURE CHARACTERISTICS

Credential	Location	Consequence of disclosure
Client configuration	Browser bundle and application package	None. Public by design; protection derives from database rules, not from concealment
Service account key	**Deployment environment only**	**Total compromise. Unrestricted read and write across all data**
Storage refresh token	**Deployment environment only**	**Full control of files created by this application**
Student password	Identity service, hashed	Access to one account until reset

The two credentials marked as total or full compromise are excluded from version control, and their absence from the repository and its complete commit history was verified before publication.

C. Acknowledged residual exposure

Accepted risk

Stored files are readable by any bearer of their address. Discovery requires an authenticated catalogue read, so the collection cannot be enumerated by an outsider. However, an address disclosed by a legitimate reader functions for any recipient, indefinitely, and cannot be revoked without deleting the file.

This is the direct cost of permitting the mobile client to retrieve content without holding a storage credential. Eliminating it requires proxying every retrieval through an authenticated endpoint, which reintroduces the bandwidth and execution duration constraints the transfer design was constructed to avoid. The trade is defensible for institutionally owned material and should be revisited before any commercially licensed title is catalogued.

D. Authorisation durability

The storage authorisation consent configuration is published rather than left in its default testing state. This distinction is consequential and easily overlooked: a configuration left in testing has its refresh credential expired by the provider after seven days [5]. Uploads then fail with an authorisation error at a point temporally remote from the configuration decision that caused it, presenting as an unexplained regression. Publication removes the expiry. No provider review was required because the narrow file scope in use is not classified as sensitive.

VII. FUNCTIONAL VERIFICATION

Fourteen test cases were executed against the deployed system rather than against a local simulation, covering authorisation boundaries, identity normalisation, the complete transfer path and delivery integrity. Results appear in Table IV. All cases passed. Records created during testing were removed on completion.

TABLE IV. FUNCTIONAL VERIFICATION RESULTS, EXECUTED 1 SEPTEMBER 2026

No.	Case	Expected behaviour	Observed
1	Administrator authentication	Session issued with administrative role assertion	Passed
2	Account creation by roll number	Account created, identifier normalised	Passed. 2024-BS-014 mapped as specified
3	Student authentication	Session issued using the issued credential	Passed
4	Role assertion in token	Token carries student role	Passed
5	Student token at administrative endpoint	Refused	Passed. Status 403
6	Unauthenticated request	Refused	Passed. Status 401
7	Duplicate identifier after normalisation	Refused as conflict	Passed. Status 409
8	Account deletion	Removed from identity service and catalogue	Passed
9	Transfer session issuance	Session address returned	Passed
10	Direct client to store transfer	Content accepted, identifier returned	Passed
11	Catalogue entry recording	Entry created with authoritative size read back from store	Passed
12	Unauthenticated retrieval	Valid file content returned	Passed. PDF signature confirmed
13	Catalogue deletion	Entry and stored file both removed	Passed. No orphaned file
14	Repository credential audit	No confidential value in working tree or history	Passed

Case 12 is the most consequential. It confirms that the retrieval path the mobile client depends upon functions without any credential, which is the property that permits the distributed package to contain no secret.

A. Interface

Figures 2 and 3 show the deployed administrative interface. Records visible were created for illustration and subsequently removed.

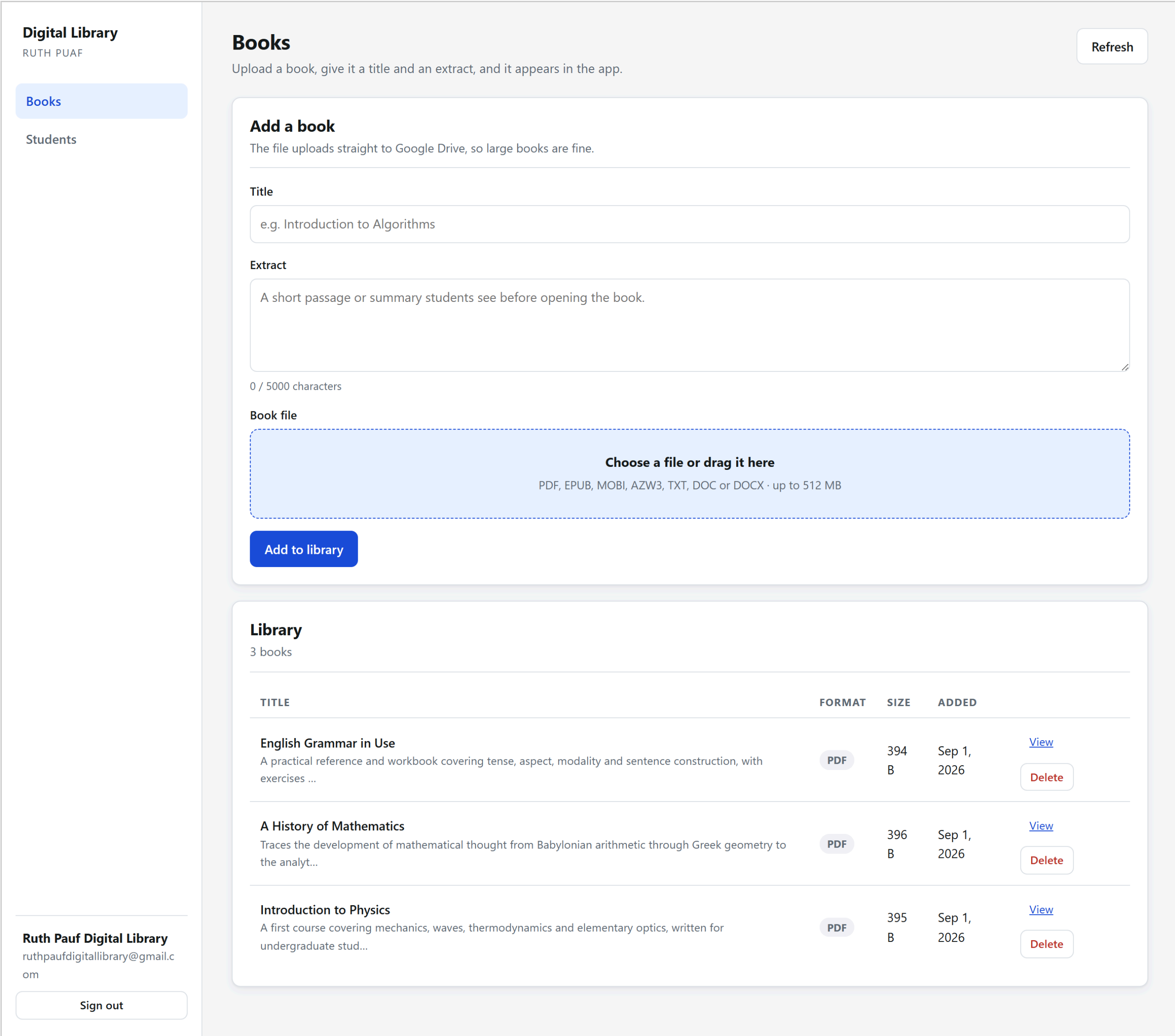

Fig. 2. Cataloguing interface. A title, a descriptive extract and a file constitute a complete record. The extract serves both as the passage shown to a reader before download and as the text searched by the mobile client, which is a coupling revisited in Section VIII.

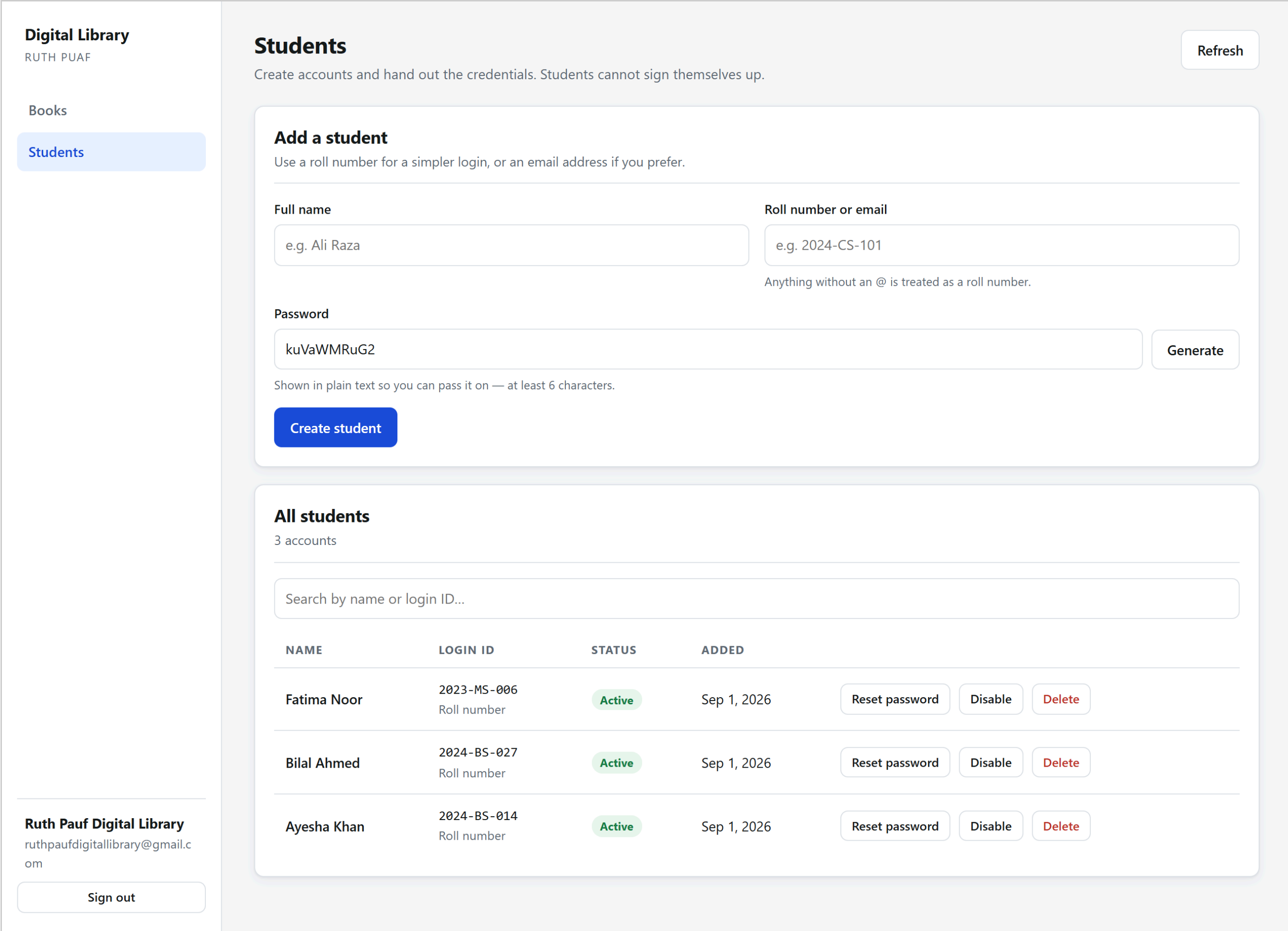

Fig. 3. Account management. Passwords are displayed once in plain text at creation because no reader initiated recovery mechanism exists. Suspension terminates active sessions immediately rather than at natural token expiry.

VIII. CAPACITY ANALYSIS

Verification establishes correctness at present scale. It says nothing about behaviour at the intended scale of two thousand to ten thousand titles. This section projects that growth and identifies where the current design ceases to hold. Three constraints bind, and their order is not the order intuition suggests.

A. Constraint 1: catalogue read volume

The mobile client subscribes to the entire catalogue collection without a page limit, and the administrative interface retrieves every record in a single request. Loading a catalogue of N titles therefore costs N document reads. With a free allowance A of fifty thousand reads per day [6], the number of complete loads L available daily is

$$L = A / N \tag{1}$$

Table V evaluates this. The result is stark. At ten thousand titles the free allowance sustains five complete catalogue loads per day across the entire student body.

TABLE V. READ COST AND FREE TIER HEADROOM AGAINST CATALOGUE SIZE

Catalogue size N	Reads per load	Loads per day L	Reads per day at 200 readers	Indicative monthly cost if metered
500	500	100	100,000	1.80 USD
2,000	2,000	25	400,000	7.20 USD
5,000	5,000	10	1,000,000	18.00 USD
10,000	**10,000**	**5**	**2,000,000**	**36.00 USD**
10,000 paginated	100	500	20,000	Within free tier

Client side persistence moderates this for repeat launches on a device holding an unchanged catalogue, which can resume from cache. It provides no relief for first installation, cleared caches or reinstallation, which is precisely the condition of a newly enrolled cohort receiving the application together.

> ***Principal finding***
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> Unbounded catalogue retrieval is the binding constraint on this design and it is reached long before any storage limit. Retrieving twenty five records per request and loading further pages on demand reduces a representative session from ten thousand reads to approximately one hundred, a reduction of two orders of magnitude, placing a ten thousand title collection comfortably within the free allowance. This single change determines whether the intended collection size is viable without recurring cost.

B. Constraint 2: cataloguing effort

This constraint is operational rather than technical and is the one most consistently underestimated. Cataloguing through the interface requires selecting a file, entering a title, composing an extract and awaiting transfer. One minute per title is an optimistic mean. Total effort T for N titles at rate r minutes per title is trivially $T = Nr$, but the consequence is not trivial.

TABLE VI. MANUAL CATALOGUING EFFORT AT ONE MINUTE PER TITLE

Catalogue size	Total hours	Working days at six productive hours	Elapsed weeks for one person
2,000 titles	33	6	1.2
5,000 titles	83	14	2.8
10,000 titles	**167**	**28**	**5.6**

Cataloguing ten thousand titles individually through a web form represents approximately six working weeks of uninterrupted effort by one person. This is not a realistic plan and would in practice be abandoned partway, leaving the collection in an inconsistent state.

The remedy is an unattended import facility accepting a directory of files alongside a tabular manifest of titles and extracts. Machine time then replaces human time for the transfer itself, and the residual human effort of preparing the manifest is divisible among several people working in parallel, which individual form entry is not.

C. Constraint 3: file storage

Storage demand depends principally on provenance. A reflowable electronic text is small; a page image scan of an older physical volume is one to two orders of magnitude larger. Table VII projects three provenance scenarios against the fifteen gigabyte free allowance, which is shared with the owning account's mail and photographs [2].

TABLE VII. PROJECTED STORAGE DEMAND BY PROVENANCE. SHADED CELLS EXCEED THE FREE ALLOWANCE

Catalogue size	Reflowable text, 2 MB mean	Born digital PDF, 8 MB mean	Page image scan, 25 MB mean
2,000 titles	4 GB	16 GB	50 GB
5,000 titles	10 GB	40 GB	125 GB
10,000 titles	20 GB	80 GB	250 GB

Only the smallest scenario fits the free allowance, and that scenario leaves no headroom for the shared consumption of mail and photographs. A paid storage tier should be treated as a certainty rather than a contingency. The determining input is the true mean file size of the intended collection, which is knowable by measurement and not by estimation: sampling one hundred representative titles and computing the mean yields a projection worth more than any figure in Table VII.

D. Constraint 4: retrieval quality

Search executes on the client by matching a substring within title and extract text. This is adequate at hundreds of titles. At ten thousand it presumes the entire catalogue is resident on the device, which is the read volume problem of Section VIII.A restated, and it cannot order results by relevance.

The catalogue database provides no full text retrieval capability. Two remediations exist. Storing a normalised term list on each record permits prefix matched queries executed by the database, which is free and adequate for title search. Integrating a dedicated retrieval service yields materially better results at the cost of an additional system to fund and maintain. The former is recommended unless retrieval within extract text proves necessary in practice.

IX. DISCUSSION AND LIMITATIONS

A. Deliberate exclusions

Lending periods, reservations and returns are absent because electronic distribution removes the scarcity that motivates them. Every active reader may hold every title simultaneously. Categories, authorship and other bibliographic metadata are absent because R6 favoured a minimal cataloguing burden; Section X proposes revisiting this once the collection is large enough that browsing by date alone becomes inadequate.

B. Distribution integrity

The distributed application package is currently signed with a development key. This is a defect with a narrow remediation window. Once a package signed with one key is installed, the platform refuses installation of an update signed with another; remediation after distribution requires every reader to uninstall, losing retrieved content. The remediation must therefore precede first distribution, and the generated key must itself be preserved, since its loss reproduces the same condition.

C. Concentration of custody

The entire collection resides in a single consumer account. Loss, suspension or compromise of that account removes the collection in its entirety. This is a severe if unlikely single point of failure, and it is inherent to the cost driven storage selection of Section III.A rather than incidental to it. An independent copy held outside that account is the proportionate mitigation.

D. Threats to validity

The projections of Section VIII rest on published service allowances and on assumed usage parameters, principally the mean file size and the frequency of complete catalogue loads. The former is measurable and should be measured before committing to a storage tier. The latter depends on client persistence behaviour under conditions not reproducible at present scale, and the read projections should therefore be treated as an upper bound rather than an expectation. The qualitative finding, that read volume binds before storage does, is robust to wide variation in both parameters.

X. FUTURE WORK

Work is ordered by the point at which deferral becomes disproportionately costly rather than by expected benefit.

Before first distribution. Generation and secure retention of a release signing key, for the reasons of Section IX.B.

Before the collection exceeds approximately one thousand titles. Pagination of catalogue retrieval in both clients, which Section VIII.A identifies as determining the viability of the intended scale. An unattended import facility, without which Section VIII.B indicates the intended scale is not reachable. Normalised term storage enabling database executed search.

As the collection matures. Bibliographic metadata supporting browsing by subject and authorship. An independent backup outside the owning account. Administrator initiated credential reset conveyed to readers. An iOS client, if a material proportion of the readership is unserved by an Android only distribution.

XI. CONCLUSION

This report has described a digital library assembled from managed services to serve an institution constrained by capital and by the absence of technical staff. The design satisfies its derived requirements at present scale, and fourteen verification cases executed against the deployed system confirm correct behaviour across authorisation boundaries, identity handling and delivery integrity.

The substantive contribution is the capacity analysis of Section VIII. It establishes that the constraint binding this design is neither storage, which intuition suggests, nor transfer size, which the architecture already resolves, but unbounded catalogue retrieval, which is reached at a collection size roughly one twentieth of that intended. It further establishes that manual cataloguing effort renders the intended scale unreachable without an unattended import facility. Both findings are quantified, both have bounded remedies, and both are substantially cheaper to address before mass cataloguing than after it.

The wider observation is that a system verified correct at demonstration scale may be structurally unsuited to its intended scale, and that the distance between the two is not apparent from functional testing alone. Capacity analysis is not a refinement to be undertaken after deployment; for this class of system it is a precondition of committing to the effort that deployment implies.

REFERENCES

1. Google, "Firebase pricing plans," Firebase Documentation. Available: https://firebase.google.com/pricing
2. Google, "Google Drive storage and Google One plans," Google Support. Available: https://support.google.com/drive
3. Vercel, "Limits for functions and requests," Vercel Documentation. Available: https://vercel.com/docs/limits
4. Google, "Perform a resumable upload," Google Drive API Documentation. Available: https://developers.google.com/workspace/drive/api/guides/manag uploads
5. Google, "Using OAuth 2.0 for web server applications: refresh token expiration," Google Identity Documentation. Available: https://developers.google.com/identity/protocols/oauth2/web-server
6. Google, "Cloud Firestore usage, quotas and pricing," Firebase Documentation. Available: https://firebase.google.com/docs/firestore/quotas
7. Google, "Cloud Firestore security rules," Firebase Documentation. Available: https://firebase.google.com/docs/firestore/security/get-started
8. Flutter, "Build and release an Android app," Flutter Documentation. Available: https://docs.flutter.dev/deployment/android

Service allowances and prices cited were current at the date of issue and vary by region. Figures 2 and 3 are screen captures of the deployed administrative interface; records shown were created for illustration and removed on completion of testing.